

TDF-Machine Learning For Prediction

Group members:

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Model1: Logistic Regression

Evaluation metrics: LR-Accuracy

Classification Report:

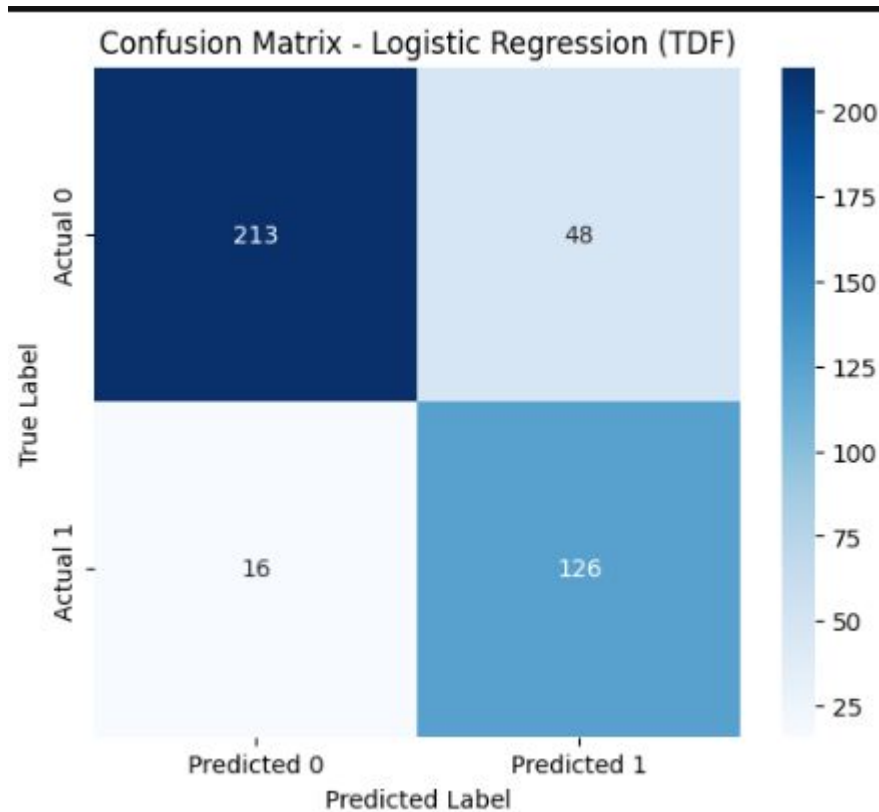
	precision	recall	f1-score	support
0	0.93	0.82	0.87	261
1	0.72	0.89	0.80	142
accuracy			0.84	403
macro avg	0.83	0.85	0.83	403
weighted avg	0.86	0.84	0.84	403

Evaluation metrics: LR-Accuracy

Classification Report:

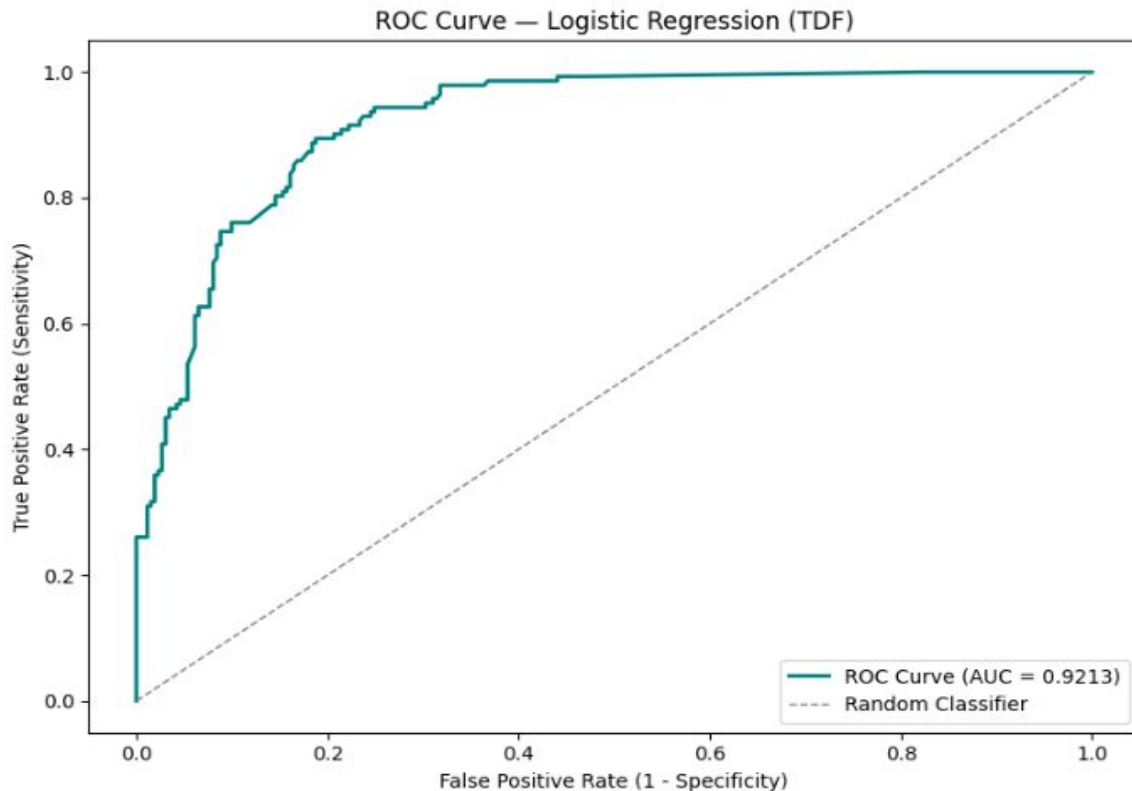
	precision	recall	f1-score	support
0	0.93	0.82	0.87	261
1	0.72	0.89	0.80	142
accuracy			0.84	403
macro avg	0.83	0.85	0.83	403
weighted avg	0.86	0.84	0.84	403

Confusion Matrix - Logistic Regression

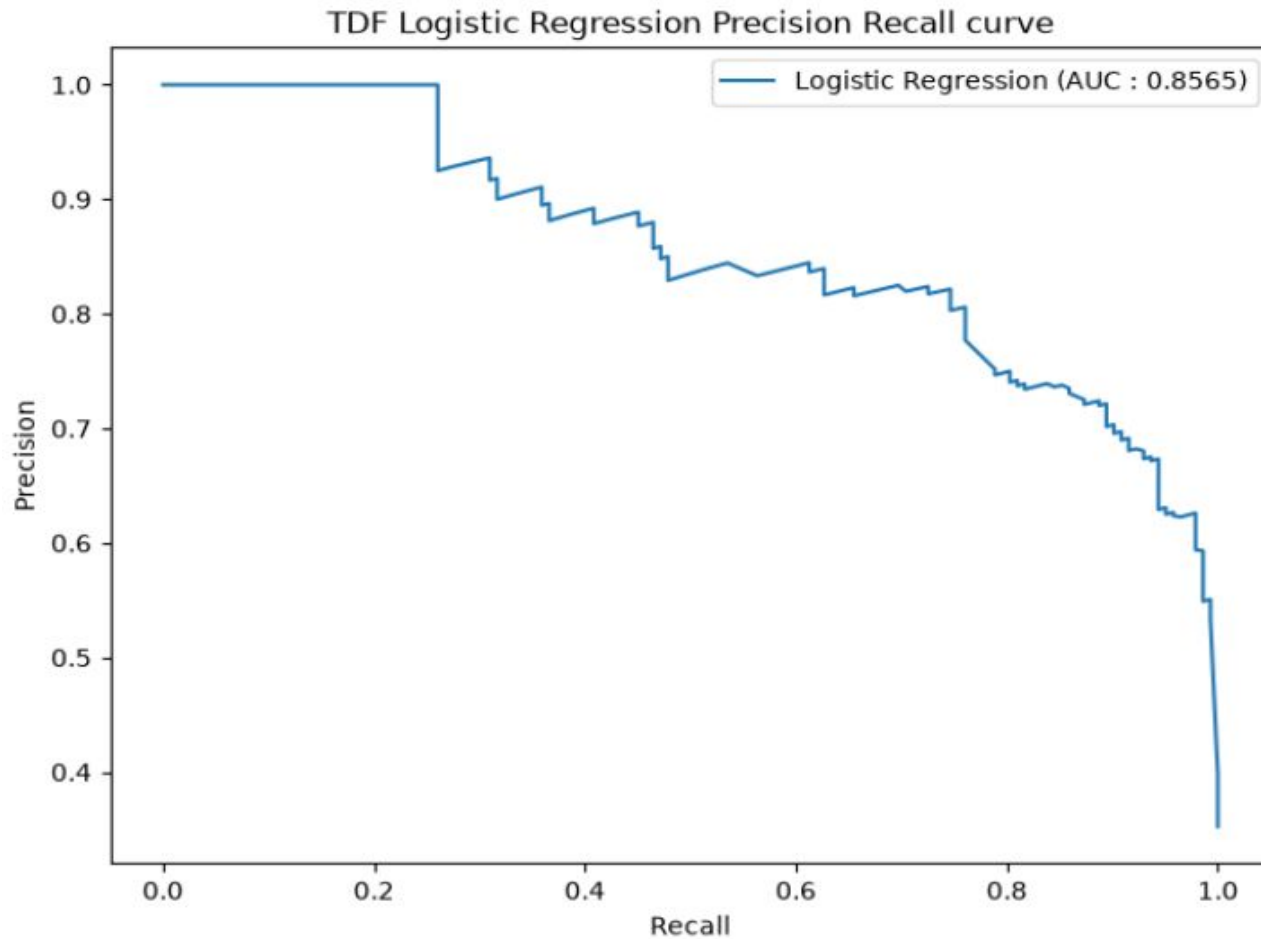


Precision, Recall, F1 score, precision- Recall & **ROC AUC curves**

- Data set is balanced - (our dataset: **Positive class = Minority**)
- False positives and False negatives are of equal importance #Critical for our model



Precision- Recall & ROC AUC curves



TDF - LR

Model 2: Random Forest Classifier

Evaluation metrics: RF-Accuracy

---- TDF Random Forest Classification Report:

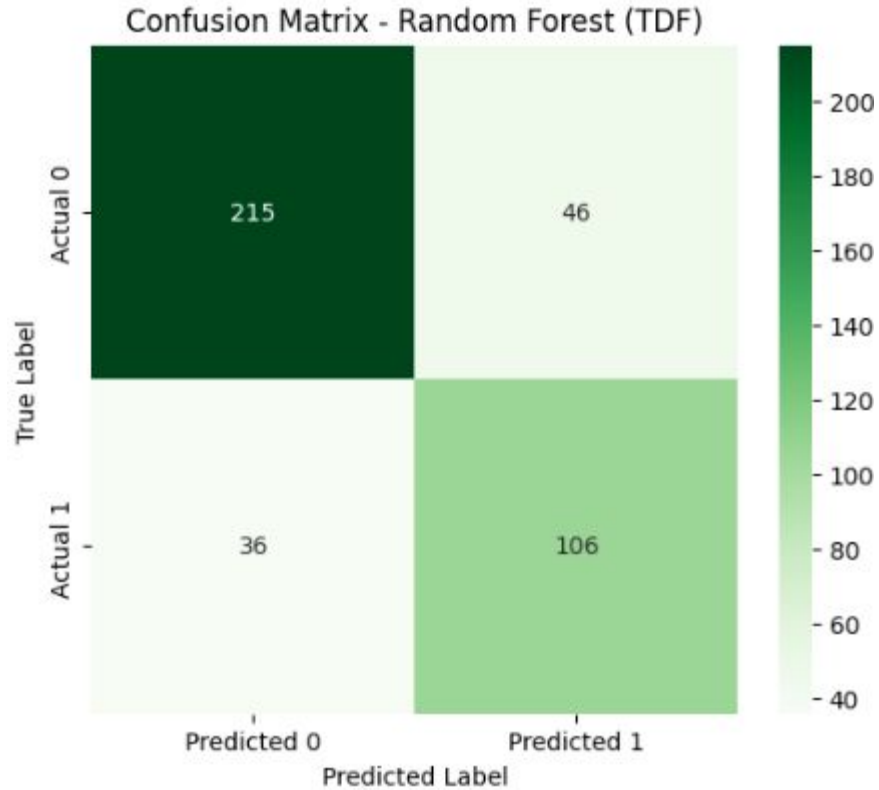
	precision	recall	f1-score	support
0	0.86	0.82	0.84	261
1	0.70	0.75	0.72	142
accuracy			0.80	403
macro avg	0.78	0.79	0.78	403
weighted avg	0.80	0.80	0.80	403

Evaluation metrics: RF-Accuracy

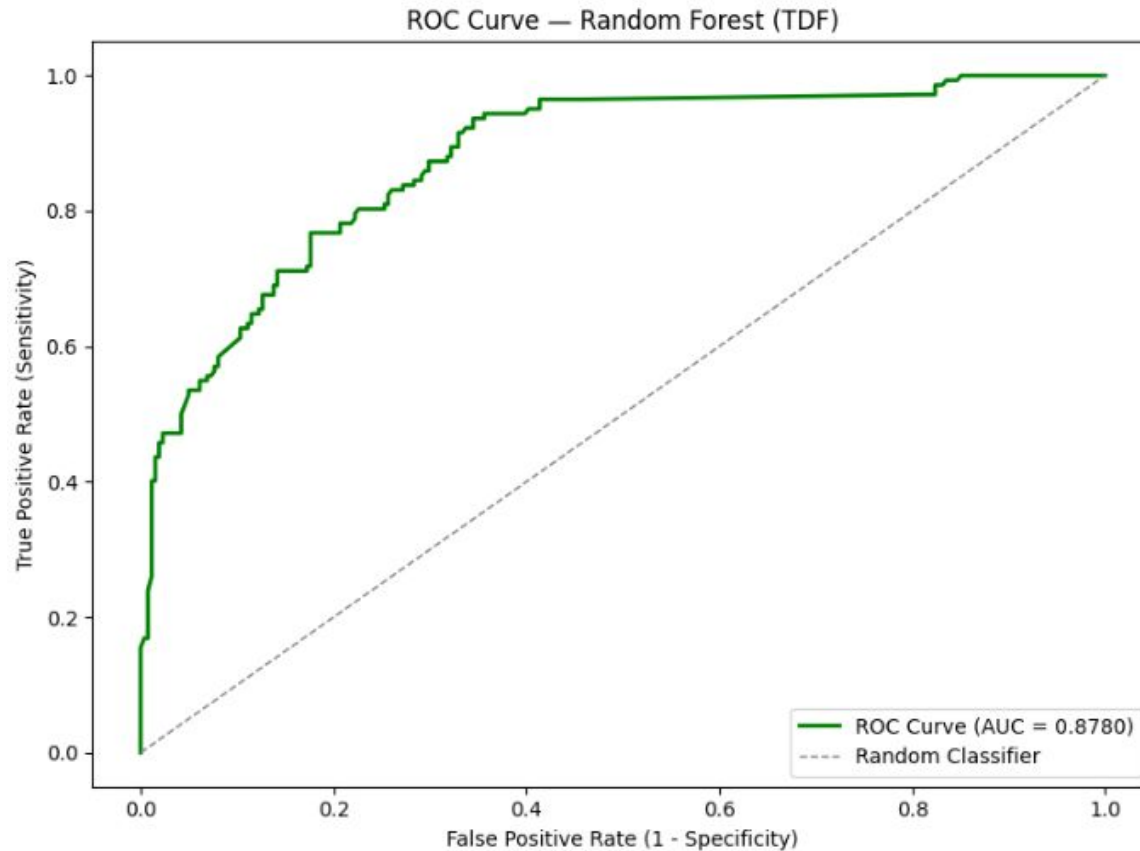
---- TDF Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.86	0.82	0.84	261
1	0.70	0.75	0.72	142
accuracy			0.80	403
macro avg	0.78	0.79	0.78	403
weighted avg	0.80	0.80	0.80	403

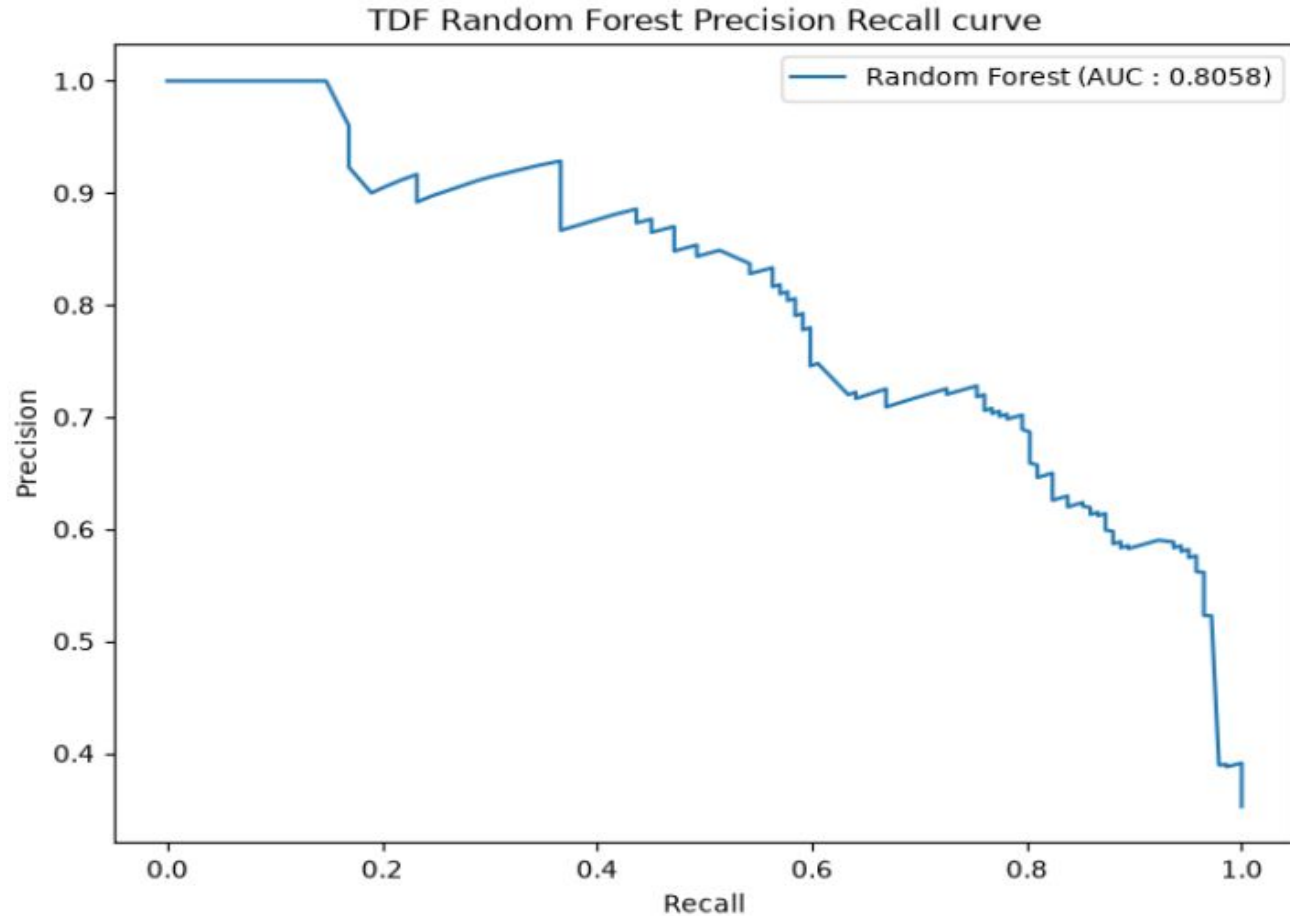
Confusion Matrix - Random Forest



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



TDF-RF

Model 3: Support Vector Machine

Evaluation metrics: SVM-Accuracy

--- SVM Test Classification Report ---

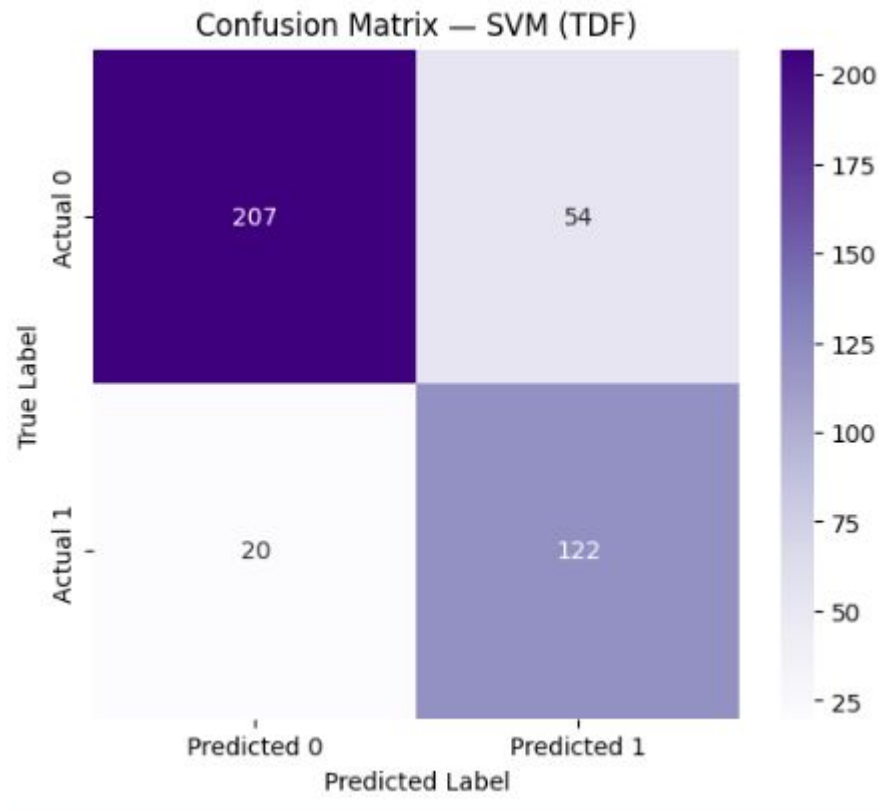
	precision	recall	f1-score	support
0	0.91	0.79	0.85	261
1	0.69	0.86	0.77	142
accuracy			0.82	403
macro avg	0.80	0.83	0.81	403
weighted avg	0.83	0.82	0.82	403

Evaluation metrics: SVM-Accuracy

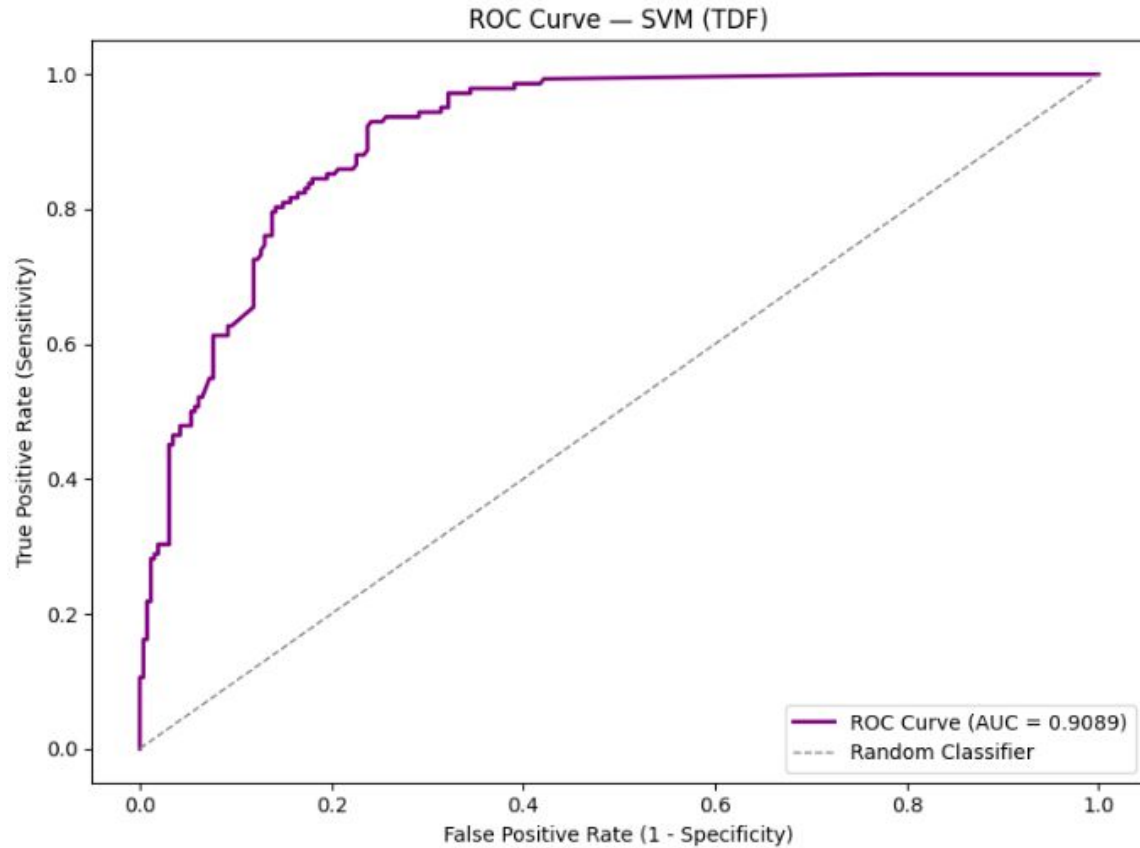
--- SVM Test Classification Report ---

	precision	recall	f1-score	support
0	0.91	0.79	0.85	261
1	0.69	0.86	0.77	142
accuracy			0.82	403
macro avg	0.80	0.83	0.81	403
weighted avg	0.83	0.82	0.82	403

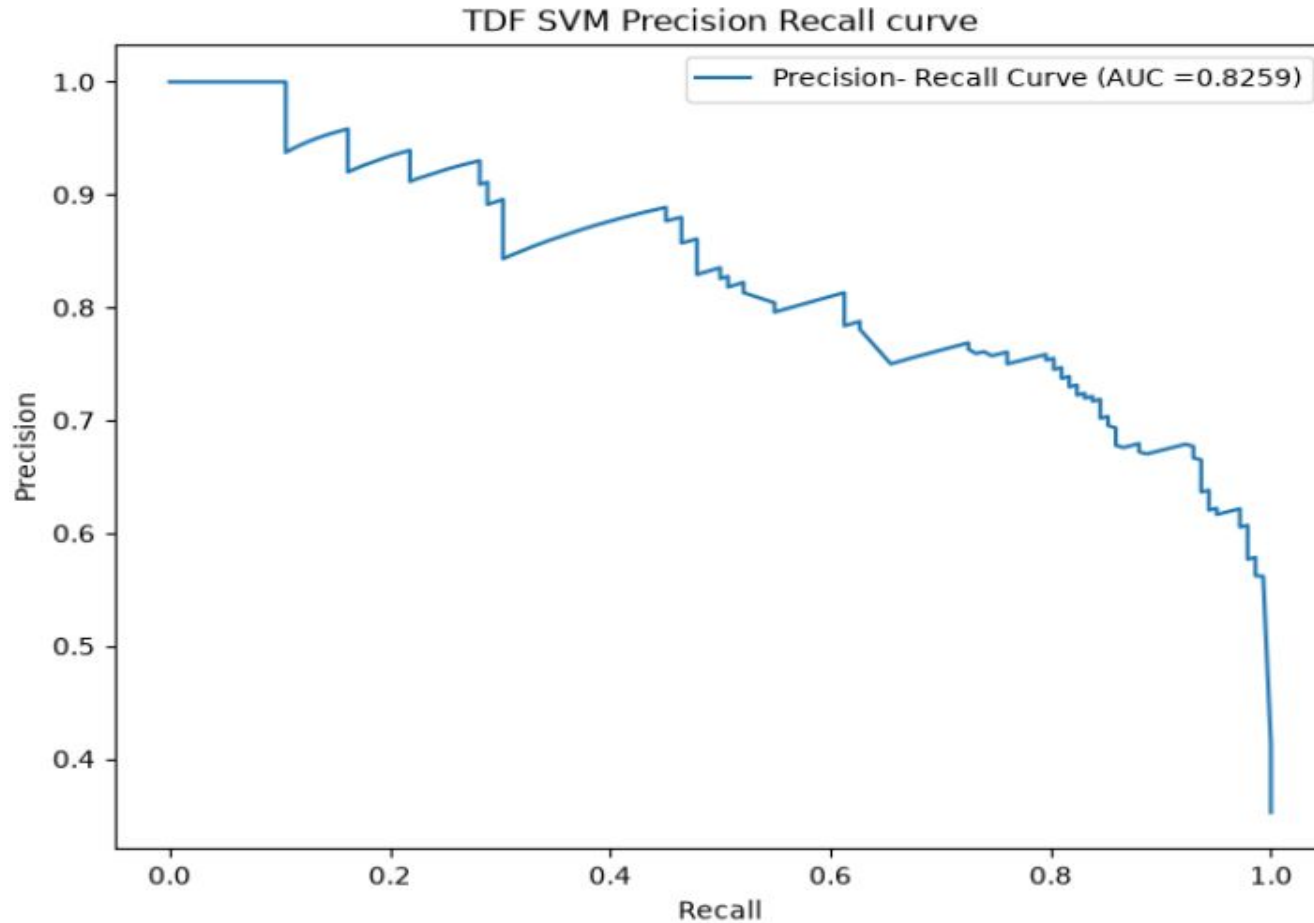
Confusion Matrix - Support Vector Machine



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



TDF-SVM

Model 4: Extreme Gradient Boost

Evaluation metrics: **XGB-Accuracy**

--- XgBoost Test Classification Report ---

	precision	recall	f1-score	support
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0	0.87	0.85	0.86	261
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1	0.73	0.76	0.75	142
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accuracy			0.82	403
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macro avg	0.80	0.81	0.80	403
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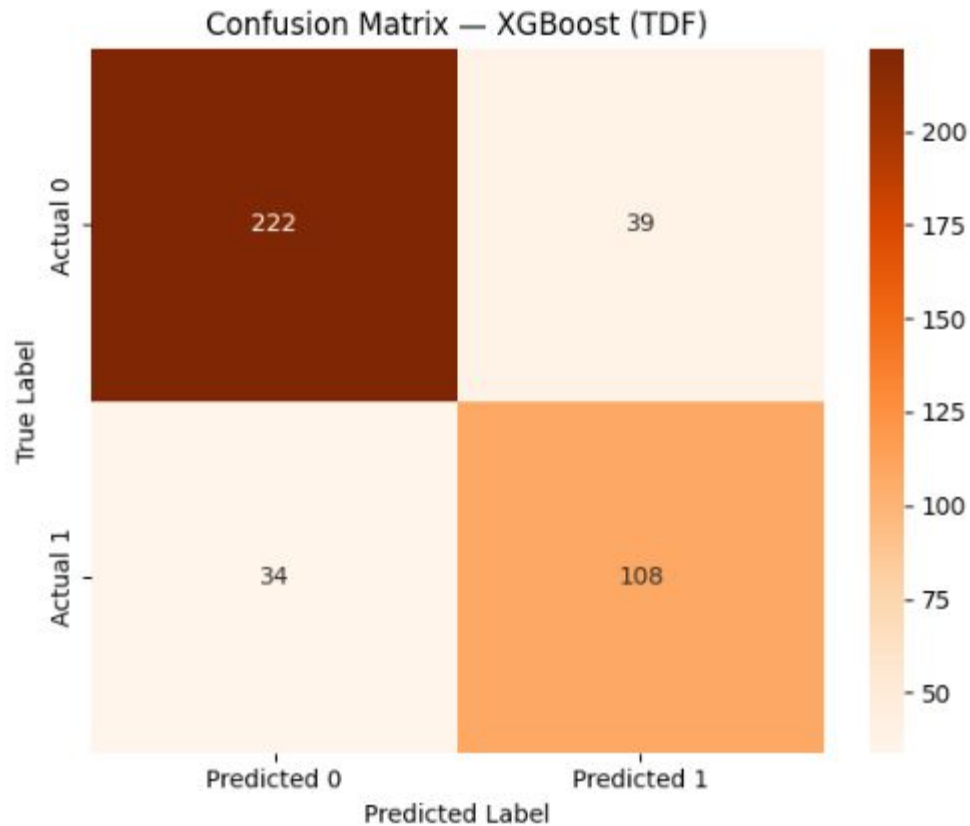
weighted avg	0.82	0.82	0.82	403
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Evaluation metrics: XGB-Accuracy

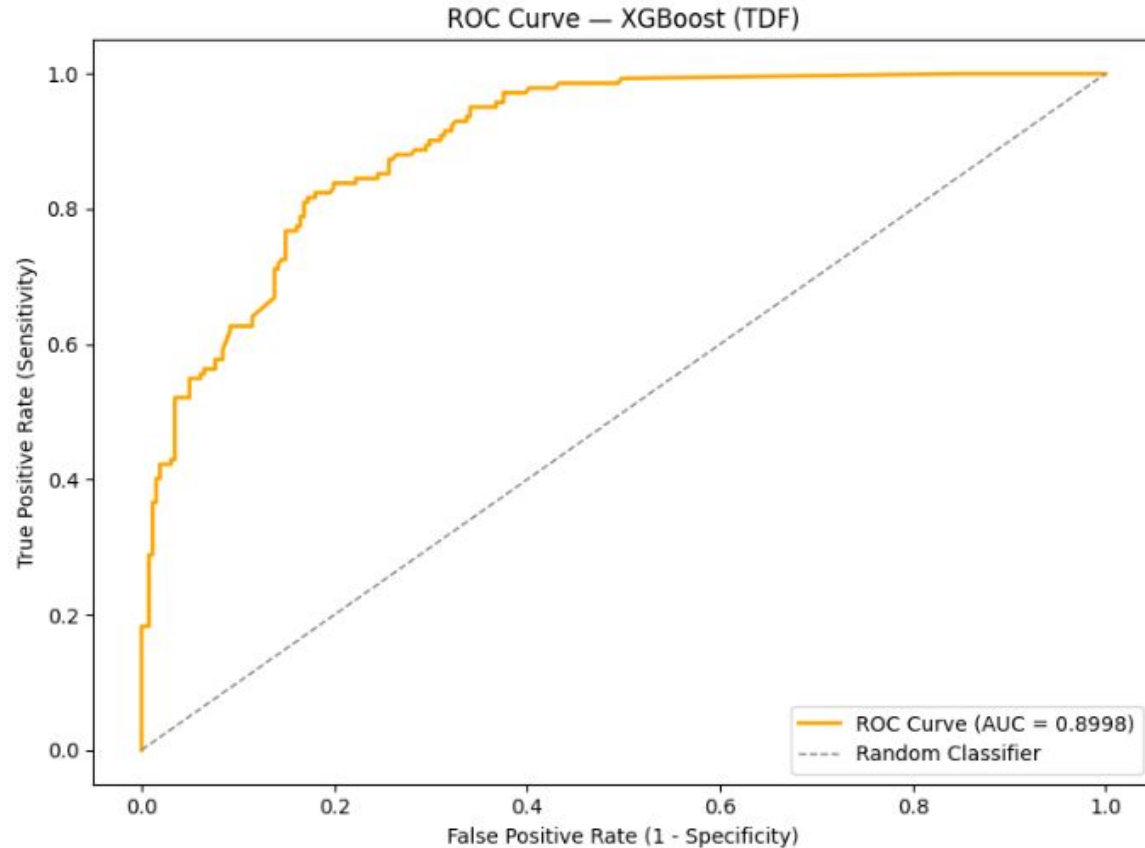
--- XgBoost Test Classification Report ---

	precision	recall	f1-score	support
0	0.87	0.85	0.86	261
1	0.73	0.76	0.75	142
accuracy			0.82	403
macro avg	0.80	0.81	0.80	403
weighted avg	0.82	0.82	0.82	403

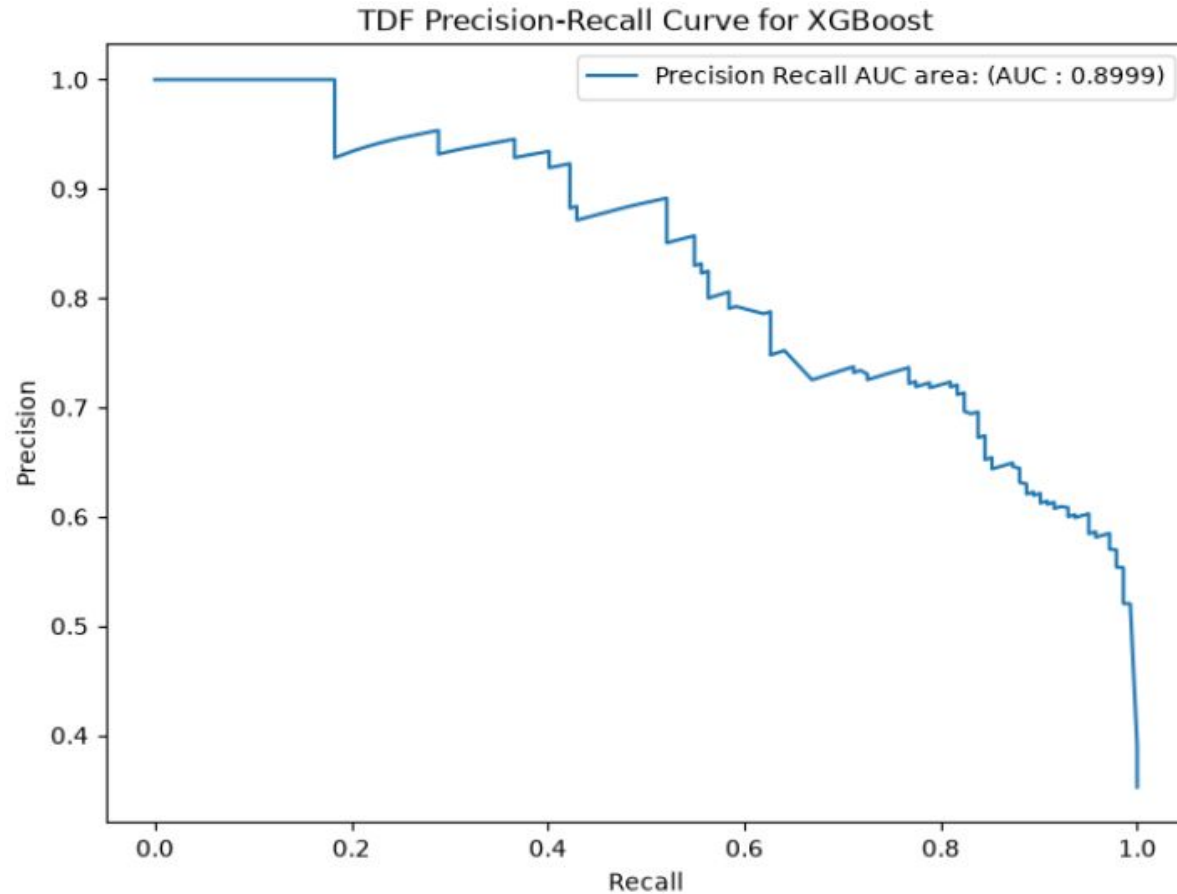
Confusion Matrix - XGBOOST



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



TDF-XGB

Comparison (Macro -avg) - TDF

Model	Precision	Recall	F1 score
LR	0.83	0.85	0.83
RF	0.78	0.79	0.78
SVM	0.80	0.83	0.81
XGB	0.80	0.81	0.80

Model Ranking by MCC (TDF)

Rank	Model	MCC
1	LR	0.6784
2	SVM	0.6282
3	XGB	0.6065
4	RF	0.5620

Best model : LR

ROC-CURVES

